FIND THE R2 VALUE IN THE MACHINE LEARNING REGRESSION

1) **Multiple Linear Regression : R_Score value = 0.78**

2) Support Vector Machine :

SNO	Hyper Parameter r value	Rbf r value	Linear r value	Poly r value	Sigmoid r value
1.	C= 0.1	-0.08	-0.12	-0.08	-0.08
2.	C=1.0	-0.08	-0.11	-0.06	-0.08
3.	C=100	-0.12	0.54	-0.09	-0.11
4.	C=1000	-0.11	0.63	-0.05	-1.66
5.	C=2000	-0.10	0.68	-0.002	-5.61
6.	C=3000	-0.09	0.75	0.04	-12.01

The **Support Vector Machine** best **R2 Value** with parameters Criterion , Max features and Splitter **= 0.75**

3) DECISION TREE :

SNo	CRITERION	MAX FEATURES	SPLITTER	R VALUE
1	squared_error	None	best	0.69
2.	squared_error	None	random	0.71
3.	squared_error	log2	best	0.48
4.	squared_error	log2	random	0.71
5.	squared_error	sqrt	best	0.76
6.	squared_error	sqrt	random	0.72
7.	friedman_mse	None	best	0.68
8.	friedman_mse	None	random	0.70
9.	friedman_mse	log2	best	0.68
10.	friedman_mse	log2	random	0.66
11.	friedman_mse	sqrt	best	0.74
12.	friedman_mse	sqrt	random	0.67
13.	absolute_error	None	best	0.65
14.	absolute_error	None	random	0.73
15.	absolute_error	log2	best	0.68
16	absolute_error	log2	random	0.72
17	absolute_error	Sqrt	best	0.69
18.	absolute_error	sqrt	Random	0.72
19	poisson	None	best	0.71
20	poisson	None	random	0.73
21	poisson	log2	best	0.73
22	poisson	log2	random	0.63
23	poisson	sqrt	best	0.73
24	poisson	sqrt	random	0.65

The **Decision Tree** best **R2 Value** with parameters Criterion , Max features and Splitter **= 0.76**

4) **Random Forest :**

SNO	CRITERION	MAX FEATURES	N_ESTIMATORS	R Value
1.	Squared_error	None	10	0.84
2.	Squared_error	None	100	0.85
3.	Squared_error	Sqrt	10	0.84
4.	Squared_error	Sqrt	100	0.87
5.	Squared_error	Log2	10	0.86
6.	Squared_error	Log2	100	0.86
7.	Absolute_error	None	10	0.83
8.	Absolute_error	none	100	0.85
9.	Absolute_error	Sqrt	10	0.86
10.	Absolute_error	Sqrt	100	0.87
11.	Absolute_error	Log2	10	0.85
12.	Absolute_error	Log2	100	0.87
13.	Friedman_mse	None	10	0.82
14.	Friedman_mse	none	100	0.85
15.	Friedman_mse	Sqrt	10	0.85
16.	Friedman_mse	Sqrt	100	0.86
17.	Friedman_mse	Log2	10	0.85
18.	Friedman_mse	Log2	100	0.86
19.	Poisson	None	10	0.85
20.	Poisson	None	100	0.85
21.	Poisson	Sqrt	10	0.85
22.	Poisson	Sqrt	100	0.86
23.	Poisson	Log2	10	0.87
24.	Poisson	Log2	100	0.86

The **Random Forest** best **R2 Value** with parameters Criterion , Max features and Splitter **= 0.87**

Questions :

1. Identify your problem statement

Dataset has both I/p and O/p clearly and it is in numerical . So this is Supervised Learning and Regression .

Goal:

To Predict Insurance Change based on the input data like age , number of children , BMI ,Sex , Smoking status

2. Tell basic info about the dataset (Total number of rows, columns)

Total No.of rows = 1338
Total No.of columns = 6

Columns :
Age (numerical)
BMI (numerical),
Children (numerical) ,
sex (categorical),
Smoker(categorical) ,
Charges(Target Variable)

3. Mention the pre-processing method if you're doing any (like converting string to number – nominal data)

I) Converted the categorical column to numerical column using One Hot Encoding

Sex => male = 1 , female = 0

Smoker => yes =1 , no = 0

Ii) Selected the independent (I/p) and dependent variable (O/p)

Input => age , bmi , children , smoker , sex

Output => Charges

Iii) Split the Training and Test Set

Training data = 80%

Test Data = 20%

4. Develop a good model with r2_score. You can use any machine learning algorithm; you can create many models. Finally, you have to come up with final model.

Used 4 Models :

* Multiple Linear Regression
* Support Vector Machine
* Decision Tree
* Random Forest

Model	R value (max)
Multiple Linear Regression	0.78
Support Vector Machine	0.75
Decision Tree	0.76
Random Forest	0.87

5) All the research values (r2_score of the models) should be documented. You can make tabulation or screenshot of the results.)

Model	Result
Multiple Linear Regression	2039.63
Support Vector Machine	1288.59
Decision Tree	1967.02
Random Forest	2301.65

6) Mention your final model, justify why u have chosen the same.

Random Forest was selected as the final modal because it has highest r2 score compared to other algorithm modals